

BASIC QUESTION

-- 1) Show books where genre is fiction

```
SELECT
    *
FROM
    books
WHERE
    genre = 'fiction';
```

-- 2) Find Books published after the year 1950

```
SELECT
    *
FROM
    books
WHERE
    Published_year > 1950;
```

-- 3) List all the customers from the canada

```
SELECT
    *
FROM
    customers
WHERE
    country = 'canada';
```

-- 4) Show order placed in November 2023

```
SELECT
    *
FROM
    orders
WHERE
```

```
order_date BETWEEN '2023-11-01' AND '2023-11-30';
```

-- 5) Retrieve total stock of books available

```
SELECT  
    SUM(stock) AS total_stock  
FROM  
    Books;
```

-- 6) Find the detail of most expensive book

```
SELECT  
    *  
FROM  
    books  
ORDER BY price DESC;
```

-- 7) Show all customer who ordered more than 1 quantity of books

```
SELECT  
    c.*, quantity  
FROM  
    customers c  
    JOIN  
    orders o ON o.customer_id = c.customer_id  
WHERE  
    quantity > 1;
```

-- 8) show all orders where total amount exceeds \$20.alter

```
SELECT  
    *  
FROM  
    orders  
WHERE
```

```
total_amount > 20;
```

-- 9) List all the genre in the books table

```
SELECT DISTINCT
```

```
    *, genre
```

```
FROM
```

```
    books;
```

-- 10) Find the book with the lowest stock

```
SELECT
```

```
    *
```

```
FROM
```

```
    books
```

```
ORDER BY stock
```

```
LIMIT 1;
```

-- 11) Calculate the total revenue generated from all orders

```
SELECT
```

```
    SUM(total_amount) AS total_revenue
```

```
FROM
```

```
    orders;
```

ADVANCED QUESTION

-- 1) Retrieve total number of books for each genre

```
SELECT
```

```
    b.genre, SUM(o.quantity) AS total_book_sold
```

```
FROM
```

```
    orders o
```

```
    LEFT JOIN
```

```
    books b ON b.book_id = o.book_id
```

```
GROUP BY genre;
```

-- 2) Find the average price of books in the "Fantasy" genre

```
SELECT
    AVG(price) AS Average_price
FROM
    books
WHERE
    genre = 'Fantasy';
```

-- 3) List customer who have placed at least 2 order

```
SELECT
    o.customer_id, c.name, COUNT(o.order_id) AS order_count
FROM
    orders o
    JOIN
    customers c ON o.customer_id = c.customer_id
GROUP BY o.customer_id , c.name
HAVING COUNT(o.order_id) >= 2;
```

-- 4) Find the most frequently ordered book

```
SELECT
    o.book_id, b.title, COUNT(o.order_id) AS order_count
FROM
    orders o
    JOIN
    books b ON o.book_id = b.book_id
GROUP BY o.book_id , b.title
ORDER BY order_count DESC;
```

-- 5) Show the top 3 most expensive books of 'Fantasy' genre;

```
SELECT
```

```

*
FROM
    books
WHERE
    genre = 'Fantasy'
ORDER BY price DESC
LIMIT 3;
```

-- 6) Retrieve total quantity of books sold by each author

```

SELECT
    b.author, SUM(o.quantity) AS total_books
FROM
    orders o
    JOIN
    books b ON o.book_id = b.book_id
GROUP BY b.author;
```

-- 7) List the cities where customer who spent over \$30 are located

```

SELECT DISTINCT
    c.city, o.total_amount
FROM
    customers c
    JOIN
    orders o ON o.customer_id = c.customer_id
WHERE
    o.total_amount > 30;
```

-- 8) Find the customers who spent the most on orders

```

SELECT
    c.name, c.customer_id, SUM(total_amount) AS total_spent
```

```
FROM
    customers c
    JOIN
        orders o ON o.customer_id = c.customer_id
GROUP BY c.customer_id , c.Name
ORDER BY total_spent DESC;
```

-- 9) Calculate the stock remaining after fulfilling all orders

```
SELECT
    b.book_id,
    b.title,
    b.stock,
    COALESCE(SUM(o.quantity), 0) AS order_quantity,
    b.stock - COALESCE(SUM(o.quantity), 0) AS remaining_quantity
FROM
    books b
    LEFT JOIN
        orders o ON b.book_id = o.book_id
GROUP BY b.book_id , b.title , b.stock;
```